
A Deterministic Judge That Cannot See Its Own Evidence

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Abstract

1 We evaluate a deterministic, environment-grounded judge that checks whether
2 an LLM’s explanation cites the sensors a graph explainer actually selected — no
3 LLM judges another LLM. On 93 traffic-forecasting scenarios it works exactly as
4 designed, yet cannot distinguish a real explanation from one whose evidence is a
5 seeded uniform random draw: precision and hallucination match to four decimals.
6 We trace this to a near-flat, unstable mask, show a downstream decision agent
7 is equally indifferent, and report a failed entropy gate. A corrector built to raise
8 grounding erases the one pre-loop measurement that did separate real evidence
9 from fake.

10 1 Introduction

11 A spatial-temporal graph neural network forecasts speed at a traffic sensor; a mask-based explainer
12 ranks which other sensors drove that forecast; an LLM turns the ranking into prose naming locations;
13 a deterministic checker verifies those locations appear in the ranking. The pipeline is *explainer* $\rightarrow$
14 *LLM* $\rightarrow$ *verifier* $\rightarrow$ *decision*. The verifier is a decidable function of the explanation artifact and a node
15 table, so it is immune by construction to the position, verbosity and self-enhancement biases afflicting
16 LLM judges (Zheng et al., 2023). That immunity buys less than it appears to. The judge correctly
17 scores citation–artifact correspondence, cannot tell whether the artifact carries any information, and a
18 correction loop added to improve grounding destroys the one signal that would have revealed the
19 evidence was fabricated.

20 2 Setup

21 The forecaster is a Graph WaveNet-style ST-GNN (Wu et al., 2019) on METR-LA (207 sensors); the
22 explainer learns a soft node mask after GNNExplainer (Ying et al., 2019), and its top- k ($k=8$) entries
23 are rendered into the prompt as the only admissible causes. The judge resolves each free-text citation
24 to a node-id set via an escalating ladder (exact sensor, exact region, fuzzy, geographic), counting a hit
25 when that set intersects the top- k ; precision, recall, F_1 and hallucination ($1 - \text{precision}$) follow. Every
26 figure we report is *macro*-averaged — the mean over scenarios of that scenario’s own value — so F_1
27 is the mean of per-scenario F_1 and, F_1 being concave in precision and recall, sits below the harmonic
28 mean of the aggregate precision and recall shown beside it. The resolver scores $29/30 = 96.7\%$
29 on hand-labeled cases — a **single-labeler validation on 30 cases**, not a reliability estimate. Five
30 conditions share one 93-scenario corpus: **A** (full explanation), **B** (prediction only), **C** (no city
31 context), **A_rand** (importances replaced by a seeded uniform draw) and **A_mismatch** (evidence
32 from a different sensor’s real explanation, via a seeded derangement). All use llama3.1:8b; every
33 call is logged, and all non-LLM stages are seeded and byte-reproducible.

Table 1: Faithfulness by condition. One 93-scenario corpus, epoch-34 checkpoint, llama3.1:8b; brackets are 95% bootstrap CIs (10,000 resamples, seed 42).

Condition	Precision	Recall	F_1	Hallucination
A (full explanation)	0.9946	0.5927	0.7246 [0.6930, 0.7553]	0.0054 [0.0000, 0.0161]
B (no explanation)	0.1720	0.0712	0.0896	0.8280
C (no city context)	0.9946	0.6048	0.7277	0.0054
A_rand (random evidence)	0.9946	0.5645	0.7013 [0.6681, 0.7334]	0.0054 [0.0000, 0.0161]
A_mismatch (wrong sensor)	0.9892	0.6008	0.7275 [0.6959, 0.7591]	0.0108 [0.0000, 0.0269]

3 Results

The judge is valid, and blind. A_rand replaces the explainer’s importances with a seeded uniform draw and changes nothing else — same scenarios, same predicted speeds, same template, prompt lengths within 0.4% of A. It scores precision 0.9946 and hallucination 0.0054, *identical to A at four decimal places*, with paired differences of exactly 0.0000 on both and 95% CIs $[-0.0161, +0.0161]$ spanning zero. A_mismatch, shown a real explanation belonging to a different sensor, reaches F_1 0.7275 against A’s 0.7246. Condition B makes this interpretable: deleting the explanation entirely does collapse the judge (precision 0.9946 $\rightarrow$ 0.1720, hallucination 0.0054 $\rightarrow$ 0.8280). The judge is not broken — it is valid for what it measures, correspondence between citations and a supplied artifact, and blind to whether that artifact means anything.

The mask is close to flat. The artifact carried little more information than noise to begin with. On an independent 207-target sweep of the same explainer, covering 80% of a target’s off-self importance mass takes on average 146.6 ± 4.7 (congested) and 154.0 ± 2.9 (free-flow) of 206 available sources.¹ A ranking needing three quarters of the network for four fifths of its mass is close to flat, and the top-8 in every prompt is the head of that flat ranking.

And its top- k is close to unstable. Flatness predicts instability, and we measure it three ways. Splitting each target’s 24 windows into two disjoint stratum-balanced halves and re-deriving the top-8 from each gives a *pooled* edge-set Jaccard — top-8 identity across the full node set, not restricted to any adjacency or reachability stratum — of 0.2256 over the 200 free-flow targets present in both halves and 0.1399 over the 28 congested ones, against a two-independent-random-draws reference of 0.0165 and 0.0182. Computed instead *per target and then averaged* — a different statistic, hence the spread — split-half agreement over 40 targets is 0.2323 ± 0.1461 at the committed sparsity coefficient, 0.2385 ± 0.1739 at 3λ and 0.2287 ± 0.1831 at 10λ : raising sparsity does not help.² Re-running the explainer at a different random seed on 10 of those targets recovers only 0.3480, 0.3149 and 0.3015 of the top-8. Even the arithmetic is load-bearing: solving the same target and window at the same seed on CPU versus MPS agrees on importance *magnitudes* (Pearson 0.9193) while agreeing on top-8 *identity* only 0.3991 of the time.³

A second explainer scores the same. On a 28-scenario subset where SHAP (KernelExplainer) was run beside the mask, the two methods’ top- k sets overlap by a mean Jaccard of 0.0221 yet receive *identical* judge precision 0.9821 and hallucination 0.0179 (F_1 0.7423 vs 0.6918).⁴

The loop erases the one signal that existed. One measurement did separate real from fabricated evidence: before any correction, 64.5% of scenarios met the 0.70 grounding threshold on real explanations against 41.9% on random ones. A corrector — score the advisory, name the top- k nodes it failed to cite, re-prompt, up to three rounds — drives both to the ceiling: real reaches 100.0% at F_1 0.8738, random 98.9% at F_1 0.8903, the fabricated condition finishing marginally *higher*

¹Run 20260824T235016Z__influence_graph_solves, metrics.json.

²Pooled figures: run 20260824T235016Z__influence_graph_solves, metrics.json, per_regime.*.stability.jaccard_all_edges — a Jaccard over the union of every target’s top-8 source–target pairs, hence no standard deviation; the adjacency-restricted variants are separate fields and are not used. Per-target figures and the entropy results below: run 20260825T201433Z__grounding_without_information, part4_metrics.json.

³Run 20260824T235643Z__device_divergence, device_divergence.json; 3 cases, Spearman rank 0.5778, top-20 Jaccard 0.3576.

⁴shap_comparison_summary.json, aggregate.{A,D} and gnn_vs_shap_topk_overlap; $n=28$.

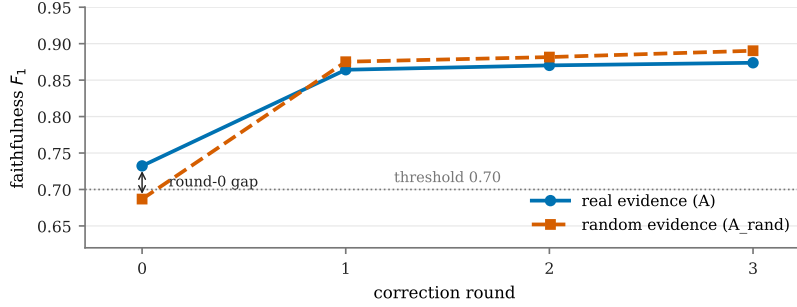


Figure 1: Faithfulness F_1 by correction round, real evidence (A) versus random evidence (A_rand), $n=93$ each, threshold 0.70. The arms are separated before any correction and indistinguishable after one round; by round 3 the fabricated condition is marginally higher. Series differ in colour, dash and marker so the figure survives greyscale.

(Figure 1). **The loop built to make the system more reliable is precisely what destroys the only measurement that revealed the evidence was fake.** What survives is agreement between prose and artifact, which the loop optimises directly; what does not is the diagnostic value of failing to agree.

3.1 The blindness reaches the decision, not just the score

A judge that cannot see its evidence would matter less if a downstream agent could. We ran the decision simulation on 150 stratified scenarios $\times$ 3 seeds, asking an agent to choose one of five traffic interventions, scored against a model-in-the-loop ground truth. Both arms are present for every scenario and seed, so the contrast is paired *within* scenario and does not involve any baseline. Showing the agent the real explanation rather than a random one changes accuracy by +0.0200 (95% CI [-0.0289, +0.0711]) and delay reduction by +0.1153 mph (95% CI [-1.3158, +1.5452]), both bootstrapped over 10,000 resamples of scenarios.⁵ **Both intervals span zero.** At the scenario level the real explanation wins 32 and loses 26 on accuracy with 92 ties, and wins 54 and loses 48 on delay reduction with 48 ties. We deliberately do not frame this against the no-explanation baseline: that baseline is unstable across sample sizes — 0.2756 accuracy on this 150-scenario subset against 0.2327 on the full 444 — by more than either explanation arm exceeds it on either, so an above-baseline test would report the sampling, not the effect.

3.2 A cheap detector that did not replicate

If the top- k is unstable because the mask is flat, then mask entropy should predict instability, and entropy is computable from a single solve whereas measuring stability costs a second one. We pre-registered that test before running it: normalised Shannon entropy of the off-self mask, correlated against the same split-half top-8 Jaccard, with success defined in advance as Spearman $\rho < -0.30$ and a bootstrap CI excluding zero, plus a threshold chosen on the sweep alone flagging all 12 committed explanations. Pooled over 120 (target, setting) pairs, $\rho = -0.2008$ with 95% CI [-0.4229, +0.0452] for the entropy of the window-averaged mask and $\rho = -0.1732$, CI [-0.3938, +0.0513] for the mean per-window entropy — the second being the only form obtainable from a single solve, and therefore the only one that would have made this cheap. Both intervals include zero; the threshold flagged 3 of 11 distinct committed explanations, not all of them.⁶ **Both pre-registered criteria are falsified and we report them as such.** A single-setting version of this test at $n=40$ did appear to pass; it did not survive the full pool, which is the ordinary fate of an underpowered result and the reason the criteria were fixed in advance.

⁵paired_decision_ab.json in the grounding run’s processed directory, computed from part3_per_decision.csv. Resampling unit is the scenario, not the decision, because a scenario’s three seeds share a window, a ground-truth action and an explanation.

⁶verdicts.json, P4; both sub-verdicts recorded FALSIFIED. The 12 committed explanation files cover 11 distinct (target, window) pairs.

99 4 Facet mapping

100 A deterministic checker’s facet coverage is settled by construction rather than measurement. *Surface-*
101 *form sensitivity*, *positional bias*, *sycophancy* and *inter-judge inconsistency* are removed outright
102 — there is no judge model to exhibit them and the same input yields the same verdict every run —
103 as is *verbosity bias*, since the score depends on the citation array, not answer length. Against that,
104 three facets are untouched. *Criteria drift* is not addressed: the evaluation field is self-declared by the
105 project that built the system, with no external specification. *Reasoning-chain validity* is not addressed:
106 the judge checks which locations are cited, never whether the causal story about them is correct, so
107 a reversed edge scores clean. And *construct validity* is what this paper puts in question — a judge
108 scoring random evidence identically to real evidence measures citation compliance, not grounding. A
109 filled disclosure template is provided in Appendix A.

110 5 Limitations

111 (a) Faithfulness here is agreement with what the model was *shown*, not with what is *true*; there is
112 no ground-truth causal attribution for a real road network, which is why A_rand is diagnostic at
113 all. (b) Condition B differs from A in several ways at once (no importance table, no propagation
114 path, different retrieved context), so we make no mechanistic claim about which absence causes its
115 collapse. (c) One claim type (citation membership), a self-declared evaluation field, and a single LLM
116 family per run: every condition in Table 1 used llama3.1:8b, and we make no cross-model claim
117 — a second model appears only in a separate cross-city study that is excluded here as a different
118 corpus. (d) Conditions A/B/C and A_rand/A_mismatch were scored by different versions of the
119 entity resolver (fixed 2026-07-21, between runs). This does not affect the precision/hallucination
120 results above — precision is identical to four decimal places between A and A_rand — since a
121 resolver-version change can only affect which citations are matched, not whether a match is correct.
122 It may affect the smaller recall/ F_1 differences between conditions; a same-corpus proxy bounds the
123 resolver’s own contribution at approximately 0.005–0.014 F_1 . We restrict our central claim to the
124 precision/hallucination axis, which is unaffected. (e) The resolver was validated on 30 hand-labeled
125 cases by a *single labeler*; there is no second annotator and no inter-annotator agreement, so 96.7%
126 is a development check, not a measured reliability. (f) The supporting measurements carry small
127 samples: seed-repeat 10 targets, device comparison 3 cases, SHAP 28 scenarios, congested split-half
128 28 targets. They are indicative; the central claim rests on Table 1. (g) The decision simulation’s
129 ground truth is model-in-the-loop — what the trained forecaster believes an intervention would do,
130 not an observed outcome — so it inherits that model’s biases and is internally consistent rather than
131 externally valid.

132 6 Related work

133 LLM-as-judge biases are set out by Zheng et al. (2023). Citation and attribution evaluation are
134 formalised by ALCE (Gao et al., 2023) and AIS (Rashkin et al., 2023), which separates attribution
135 from correctness by definition; our results instantiate that separation where no gold reference exists.
136 Shi et al. (2023) show irrelevant context degrades task accuracy; here the irrelevant content is the
137 evidence the model is told to cite.

138 The stability results sit on existing work on GNN-explainer evaluation: Sanchez-Lengeling et al.
139 (2020) benchmark attribution against computable ground truth on synthetic graphs and Agarwal et
140 al. (2023) supply generators and a library for the same purpose, while Faber et al. (2021) show that
141 comparing explanations to ground truth is itself unsound when the GNN does not use the ground-truth
142 edges, separating responsible, causing and explanatory motifs. Our setting has no ground truth at
143 all — which is what makes a randomised-evidence control necessary rather than optional — and our
144 mismatch lies one step further downstream, between an explanation and the judge that consumes it;
145 what we add is the consequence, that an unstable, near-flat attribution is indistinguishable from noise
146 to that judge and to the agent built on it. To our knowledge this combination — *GNN-explanation*
147 *artifact* as cited source, *deterministic* verifier, no gold reference, and a *correction loop* measured
148 against noise — has not been evaluated before.⁷

⁷C_RICH (fuller context) and D_CONTRA (contradictory context) ran on this corpus in a separate LLM draw and did not move hallucination materially; being a different draw they are omitted from Table 1.

References

- C. Agarwal, O. Queen, H. Lakkaraju, and M. Zitnik. Evaluating explainability for graph neural networks. *Scientific Data*, 10:144, 2023.
- L. Faber, A. K. Moghaddam, and R. Wattenhofer. When comparing to ground truth is wrong: On evaluating GNN explanation methods. In *KDD*, 2021.
- T. Gao, H. Yen, J. Yu, and D. Chen. Enabling large language models to generate text with citations. In *EMNLP*, 2023. arXiv:2305.14627.
- H. Rashkin, V. Nikolaev, M. Lamm, L. Aroyo, M. Collins, D. Das, S. Petrov, G. S. Tomar, I. Turc, and D. Reitter. Measuring attribution in natural language generation models. *Computational Linguistics*, 49(4):777–840, 2023.
- B. Sanchez-Lengeling, J. N. Wei, B. K. Lee, E. Reif, P. Wang, W. W. Qian, K. McCloskey, L. J. Colwell, and A. B. Wiltschko. Evaluating attribution for graph neural networks. In *NeurIPS*, 2020.
- F. Shi, X. Chen, K. Misra, N. Scales, D. Dohan, E. H. Chi, N. Schärli, and D. Zhou. Large language models can be easily distracted by irrelevant context. In *ICML*, 2023. arXiv:2302.00093.
- Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang. Graph WaveNet for deep spatial-temporal graph modeling. In *IJCAI*, pages 1907–1913, 2019.
- R. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec. GNNExplainer: Generating explanations for graph neural networks. In *NeurIPS*, 2019. arXiv:1903.03894.
- L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *NeurIPS Datasets and Benchmarks*, 2023. arXiv:2306.05685.

A JUDGE Judge Deployment Disclosure Template (filled)

- Judge identity.** Deterministic program, not a model. Entity resolver plus set-intersection scoring over the explainer’s top- k . No LLM is used to judge another LLM’s output. Version-sensitive: the resolver was changed 2026-07-21.
- What is judged.** A single claim type: whether a free-text location in the LLM’s `cited_causes` [] array resolves to a node set intersecting the explainer’s top- k . The LLM’s prose is not read. A refusal (no citations) scores hallucination 1.0, identical to confidently inventing causes.
- Scale and outputs.** Precision, recall, F_1 , hallucination rate ($1 - \text{precision}$), per advisory, averaged over 93 scenarios.
- Ground truth.** None available. The reference is the explainer’s own top- k , so the judge measures correspondence, not correctness.
- Validation.** Resolver: $29/30 = 96.7\%$ on hand-labeled cases, one labeler, no inter-annotator agreement, threshold set at 90% in advance.
- Known blind spots.** Cannot distinguish real from random evidence (Table 1); cannot distinguish overreach from falsehood, since both score as $1 - \text{precision}$; does not read the reasoning chain; scores a refusal identically to a fabrication.
- Criteria provenance.** Self-declared by the authors of the system being evaluated. Not externally specified, not community-standardised.
- Reproducibility.** All non-LLM stages seeded and byte-reproducible. LLM calls are logged in full but are *not* byte-reproducible: they run at temperature 0.1 on an unseeded sampling path, chosen so results remain comparable with previously committed numbers produced on that path.

191 B Claim-to-source map

192 Every number in the main text traces to one of the following. Run
 193 directories are under evaluation/results/; GWI abbreviates run
 194 20260825T201433Z__grounding_without_information.

Claim	Value	Source
A / B / C rows	Table 1	claims.csv ← faithfulness_summary.json
A_rand / A_mismatch rows	Table 1	claims.csv ← GWI part1_summary.json
Bootstrap CIs, paired diffs	10k, seed 42	GWI part1_summary.json, bootstrap_ci
Corpus, epoch 34, $n=93$	2026-07-05/06	claims_provenance.json, corpus
Prompt lengths within 0.4%	+0.18%, +0.38%	GWI part1_token_parity.json
80% mass of importance	146.6 ± 4.7 / 154.0 ± 2.9 of 206	20260824T235016Z__influence_ graph_solves, metrics.json
Split-half J, <i>pooled</i> (200 / 28 targets)	0.2256 / 0.1399; ref 0.0165 / 0.0182	same file, per_regime.*.stability. jaccard_all_edges
SHAP overlap and scores	J 0.0221; F_1 0.7423 / 0.6918; P 0.9821	shap_comparison_summary.json
195 Split-half J, <i>per-target mean</i> , 3 λ	0.2323 / 0.2385 / 0.2287	GWI part4_metrics.json, settings[]
Seed-repeat Jaccard, 3 λ	0.3480 / 0.3149 / 0.3015 ($n=10$)	same file
CPU vs MPS divergence	Pearson 0.9193, top-8 J 0.3991	20260824T235643Z__device_ divergence, device_divergence.json
Round-0 grounding	64.5% real / 41.9% random	active_grounding_summary.json; GWI part2_summary.json
Loop curve, Figure 1	real F_1 0.7323→0.8738; random 0.6868→0.8903	same two files, per_round[]
Loop reach endpoints	100.0% / 98.9%	same two files
Paired decision contrast	+0.0200; +0.1153 mph	GWI paired_decision_ab.json
Baseline instability	0.2756 ($n=150$) vs 0.2327 ($n=444$)	same file, raw_baseline_instability
Entropy gate ρ and CIs	−0.2008, −0.1732; 3/11 flagged	GWI verdicts.json, P4
Resolver 29/30 = 96.7%	threshold 0.90	resolver_accuracy.json
Resolver proxy 0.005–0.014 F_1	same-corpus bound	claims_provenance.json